

Credit Risk Measurement
Advanced Topics in Banking - Project 2
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1 Task 1 — One-Year Migration Matrix (Lando–Skodeberg)

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1.1 Case-Study Task

Based on the rating history of 10,000 representative borrowers, determine the one-year migration matrix based on the Lando and Skodeberg procedure. Determine the migration matrix using the generator matrix based on at least the first five terms of the matrix exponential.

1.2 Methodology

The Lando–Skodeberg [2002] estimator models rating dynamics as a time-homogeneous Markov chain in continuous time. The generator matrix Λ collects the instantaneous transition intensities. Its maximum-likelihood estimator, given the full rating history of all obligors, is

$$\hat{\lambda}_{ij} = \frac{N_{ij}}{\int_0^T Y_i(s) ds}, \quad i \neq j, \quad \hat{\lambda}_{ii} = -\sum_{j \neq i} \hat{\lambda}_{ij},$$

where N_{ij} is the observed number of migrations from rating i to rating j during the window $[0, T]$, and $Y_i(s)$ is the number of obligors in state i at time s , so the denominator is the total obligor-time spent in the from-state (slide p.126 of docs/01_Slides_QFin_AT in Banking_V1_Handout.pdf). The one-year migration matrix is then obtained by exponentiating the generator,

$$P(\Delta t) = e^{\Lambda \cdot \Delta t} = \sum_{k=0}^{\infty} \frac{(\Lambda \cdot \Delta t)^k}{k!},$$

which we evaluate via a truncated Taylor expansion (slide p.132). Default is modelled as an absorbing state, so the row of Λ corresponding to D is identically zero.

1.3 State Space and Long-Format Transitions

The 8-state rating universe used in this report is ordered from highest quality to default, matching the lecture convention:

```
ratings_order <- c("AAA", "AA", "A", "BBB", "BB", "B", "CCC/C", "D")
K <- length(ratings_order)

# Wide -> long: one row per (Obligor, snapshot date, rating).
ratings_long <- melt(
  ratings,
```

```

id.vars      = "Obligor",
variable.name = "date",
value.name   = "rating",
variable.factor = FALSE
)
ratings_long[, date := as.Date(date)]
ratings_long[, rating := factor(rating, levels = ratings_order)]
setorder(ratings_long, Obligor, date)

# Add the previous-month rating within each obligor; the first row of each
# obligor receives NA because no prior snapshot exists.
ratings_long[, prev_rating := shift(rating, n = 1L, type = "lag"),
              by = Obligor]
transitions <- ratings_long[!is.na(prev_rating)]

```

Table 1: Sample of obligor transitions (first five rows of the long table).

	Obligor	date	prev_rating	rating
	1	2025-01-31	CCC/C	CCC/C
	1	2025-02-28	CCC/C	CCC/C
	1	2025-03-31	CCC/C	B
	1	2025-04-30	B	B
	1	2025-05-31	B	B

Reshaping from wide to long yields one row per obligor-snapshot observation, and lagging within Obligor produces the (`prev_rating` -> `rating`) pairs that feed the migration count. Pairs where `prev_rating` is NA (the first observation of each obligor) carry no transition information and are dropped. The data span 13 monthly snapshots from 2024-12-31 to 2025-12-31, giving 12 transitions per obligor and a total of $10,000 \times 12 = 120,000$ transition pairs.

1.4 Migration Counts N_{ij}

```

N <- unclass(table(from = transitions$prev_rating,
                  to   = transitions$rating))
storage.mode(N) <- "double"

```

Table 2: Migration count matrix N_{ij} : observed transitions from row-rating to column-rating.

	AAA	AA	A	BBB	BB	B	CCC/C	D
AAA	5,860	37	4	1	1	0	0	0
AA	7	11,417	85	11	0	0	0	0
A	1	39	23,191	104	6	4	1	4
BBB	1	11	120	35,982	164	28	7	10
BB	2	0	5	102	23,440	163	22	19
B	0	0	1	2	52	11,927	62	65
CCC/C	0	0	4	6	7	45	5,322	161

D	0	0	0	0	0	0	0	1,497
---	---	---	---	---	---	---	---	-------

The count matrix is strongly diagonal-dominant: obligors are far more likely to stay in their current rating than to migrate. Off-diagonal mass concentrates immediately adjacent to the diagonal — single-notch moves dominate two-notch moves — consistent with the rating-momentum literature. The bottom row is identically zero off the diagonal because default is absorbing in this state space, so no transitions originate from D. The first few new defaults (4 from A, 10 from BBB, 19 from BB, 65 from B, 161 from CCC/C) are visible in the last column and rise monotonically with risk, which is the qualitative behaviour the IRB and simulation calibrations will inherit.

1.5 Time in State Y_i

```
# Add previous snapshot date alongside previous rating, then compute the
# length of each interval (in years) for the from-state's account.
ratings_long[, prev_date := shift(date, n = 1L, type = "lag"),
              by = Obligor]
ratings_long[, dt := as.numeric(date - prev_date) / 365.25]

Y <- tapply(ratings_long$dt, ratings_long$prev_rating,
            sum, na.rm = TRUE)
Y <- Y[ratings_order]
```

Table 3: Total time spent by all obligors in each rating i (obligor-years).

Rating	Obligor-years
AAA	491.49
AA	959.33
A	1944.64
BBB	3024.57
BB	1977.84
B	1008.54
CCC/C	461.51
D	125.23

Each obligor contributes the year-fraction between consecutive snapshots to the rating it occupied at the start of that interval. Summing across obligors gives the denominator of the Lando–Skodeberg estimator. The exposure-time distribution mirrors the data’s rating distribution: the investment-grade buckets (BBB, A, BB) accumulate the most obligor-years, while AAA and CCC/C are sparsely populated, which mechanically inflates the variance of those rows of Λ . The total $\sum_i Y_i \approx 9,993$ obligor-years almost exactly equals $10,000$ obligors $\times$ 1 year of window; the small shortfall reflects the unequal calendar lengths of individual months.

1.6 Generator Matrix Λ

```
# Off-diagonal entries:  $\lambda_{ij} = N_{ij} / Y_i$  (sweep divides row  $i$  by  $Y_i$ ).
Lambda <- sweep(N, MARGIN = 1, STATS = Y, FUN = "/")
```

```
# Stays are not transitions: wipe the diagonal, then set it to the negative
# row-sum so each row of Lambda sums to zero.
diag(Lambda) <- 0
diag(Lambda) <- -rowSums(Lambda)
```

Table 4: Estimated generator Λ (annualised transition intensities).

	AAA	AA	A	BBB	BB	B	CCC/C	D
AAA	-0.0875	0.0753	0.0081	0.0020	0.0020	0.0000	0.0000	0.0000
AA	0.0073	-0.1074	0.0886	0.0115	0.0000	0.0000	0.0000	0.0000
A	0.0005	0.0201	-0.0818	0.0535	0.0031	0.0021	0.0005	0.0021
BBB	0.0003	0.0036	0.0397	-0.1127	0.0542	0.0093	0.0023	0.0033
BB	0.0010	0.0000	0.0025	0.0516	-0.1583	0.0824	0.0111	0.0096
B	0.0000	0.0000	0.0010	0.0020	0.0516	-0.1805	0.0615	0.0644
CCC/C	0.0000	0.0000	0.0087	0.0130	0.0152	0.0975	-0.4832	0.3489
D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Off-diagonal entries are instantaneous transition intensities, while the diagonal equals the negative row-sum so mass is conserved. Row sums are zero to machine precision (largest absolute deviation $1.1\text{e-}17$). Intensities grow sharply as we move down the rating ladder: the magnitude of the CCC/C diagonal is roughly an order of magnitude larger than the AAA diagonal, meaning CCC/C-rated obligors leave their bucket much faster than high-grade names — which, after exponentiation, will translate into PDs differing by several orders of magnitude.

1.7 Matrix Exponential — One-Year Migration Matrix

We implement the truncated Taylor expansion of e^Λ with the standard recursion $\text{term}_k = \text{term}_{k-1} \cdot \Lambda/k$, which avoids any explicit factorial and uses one matrix multiplication per iteration:

```
expm_taylor <- function(M, k_max) {
  K <- nrow(M)
  P <- diag(K)           # k = 0 term (identity)
  term <- diag(K)         # running term M^k / k!
  for (k in seq_len(k_max)) {
    term <- (term %*% M) / k
    P <- P + term
  }
  dimnames(P) <- dimnames(M)
  P
}

P5 <- expm_taylor(Lambda, k_max = 5)   # case-study minimum
P20 <- expm_taylor(Lambda, k_max = 20) # essentially exact

trunc_error <- max(abs(P5 - P20))
expm_compare <- max(abs(P20 - expm::expm(Lambda)))
```

Table 5: One-year migration matrix $P^{(5)}$ from the 5-term Taylor expansion (%).

	AAA	AA	A	BBB	BB	B	CCC/C	D
AAA	91.648	6.840	1.057	0.253	0.188	0.010	0.002	0.003
AA	0.665	89.927	8.090	1.244	0.044	0.015	0.003	0.011
A	0.055	1.837	92.329	4.876	0.409	0.220	0.052	0.224
BBB	0.035	0.363	3.626	89.562	4.771	1.007	0.223	0.412
BB	0.090	0.015	0.325	4.531	85.674	7.035	1.011	1.319
B	0.002	0.002	0.119	0.321	4.403	83.898	4.459	6.796
CCC/C	0.001	0.009	0.686	1.033	1.329	7.091	61.893	27.958
D	0.000	0.000	0.000	0.000	0.000	0.000	0.000	100.000

Table 6: One-year migration matrix $P^{(20)}$ from the 20-term Taylor expansion (%).

	AAA	AA	A	BBB	BB	B	CCC/C	D
AAA	91.648	6.840	1.057	0.253	0.188	0.010	0.002	0.003
AA	0.665	89.927	8.090	1.244	0.044	0.015	0.003	0.011
A	0.055	1.837	92.329	4.876	0.409	0.220	0.052	0.224
BBB	0.035	0.363	3.626	89.562	4.771	1.007	0.223	0.412
BB	0.090	0.015	0.325	4.531	85.674	7.035	1.011	1.319
B	0.002	0.002	0.119	0.321	4.403	83.898	4.459	6.796
CCC/C	0.001	0.009	0.686	1.033	1.329	7.091	61.895	27.956
D	0.000	0.000	0.000	0.000	0.000	0.000	0.000	100.000

The case-study requires at least five terms of the series (slide p.132); we additionally compute the 20-term expansion as a benchmark. The maximum absolute element-wise difference between $P^{(5)}$ and $P^{(20)}$ is $2e-05$, i.e. the 5-term truncation is already accurate to about two parts in 10^5 on every cell. For external validation, $P^{(20)}$ agrees with `expm::expm(Lambda)` to $2.2e-16$, which is machine precision. The diagonal entries of $P^{(20)}$ give the one-year *persistence* probability for each rating, falling from \$ 91.6% for AAA to \$ 61.9% for CCC/C, and the last row equals $(0, \dots, 0, 1)$ exactly, confirming that default remains absorbing under the exponential map.

1.8 One-Year PD Vector (with 5 bp Floor)

```
PD_raw <- P20[, "D"]
PD      <- pmax(PD_raw, 0.0005) # 5 basis-point regulatory floor
```

Table 7: One-year probabilities of default by rating, before and after the 5 bp regulatory floor.

Rating	PD_raw	PD	Floored
AAA	0.000027	0.000500	TRUE
AA	0.000114	0.000500	TRUE
A	0.002237	0.002237	FALSE
BBB	0.004118	0.004118	FALSE
BB	0.013191	0.013191	FALSE

B	0.067963	0.067963	FALSE
CCC/C	0.279564	0.279564	FALSE
D	1.000000	1.000000	FALSE

The raw AAA and AA PDs (approximately $2.7\text{e-}05$ and $1.1\text{e-}04$) lie below the 5 basis-point floor required by the regulator (CRR Art. 160(1)). This is a well-known feature of investment-grade migration data: the highest-quality buckets simply do not generate enough defaults in a finite sample for the empirical estimator to be reliable. The floor is the supervisor's way of acknowledging that even AAA names carry strictly positive default risk. From A downward the empirical PD dominates the floor by at least three orders of magnitude, with CCC/C reaching 28%. The full ordering $\text{PD}_{\text{AAA}} < \text{PD}_{\text{AA}} < \dots < \text{PD}_{\text{CCC/C}}$ is preserved, which is necessary for the downstream regulatory and simulation models to behave sensibly.

1.9 Summary — Inputs for Downstream Tasks

The three objects carried forward into Tasks 3, 4 and 5 are:

- **PD** — 8-vector of one-year PDs by rating, after the 5 bp floor. Primary input to the IRB closed-form formula (Task 3) and to both Merton-style simulations (Tasks 4 and 5).
- **P20** — 8×8 one-year migration matrix. Useful if a downstream task needs the full migration probabilities rather than just the default column.
- **Lambda** — the underlying generator. Useful for computing migration probabilities at any horizon Δt via $P(\Delta t) = e^{\Lambda \Delta t}$.

2 Task 2 — Basel Standardized Approach

Author: Farkas Tallos

2.1 Case-Study Task

Determine the credit risk for the portfolio of 100 borrowers based on the Basel Standardized Approach. Proceed by using the existing ratings as if they were external ratings.

2.2 Regulatory Risk-Weight Lookup (CRR Art. 122)

Under the Standardized Approach the risk weight applied to a corporate exposure is a step function of the external credit assessment (CRR Article 122; slide p.208 of `docs/01_Slides_QFin_AT in Banking_V1_Handout.pdf`). Following the case-study instruction, we treat the internal ratings as if they were ECAI ratings. The mapping from the eight-bucket rating universe to risk weights, with defaulted exposures attracting the 150% weight applicable when specific provisions cover less than 20% of the outstanding amount, is:

```
rw_table <- data.table(
  Rating = c("AAA", "AA", "A", "BBB", "BB", "B", "CCC/C", "D"),
  RW     = c(0.20, 0.20, 0.50, 1.00, 1.00, 1.50, 1.50, 1.50)
)
```

Table 8: CRR Art. 122 risk weights for corporate exposures.

Rating	Risk Weight
AAA	0.2
AA	0.2
A	0.5
BBB	1.0
BB	1.0
B	1.5
CCC/C	1.5
D	1.5

2.3 Per-Obligor Risk-Weighted Assets

```
port_sa <- merge(portfolio, rw_table, by = "Rating", all.x = TRUE)
port_sa[, RWA := Loan.Volume * RW]
port_sa[, Capital := 0.08 * RWA]
setorder(port_sa, -RWA)
```

Table 9: Ten largest obligors by RWA under the Standardized Approach.

Obligor	Rating	Industry	Loan.Volume	RW	RWA	Capital
27	CCC/C	1	850,000	1.5	1,275,000	102,000
3	BB	2	850,000	1.0	850,000	68,000
21	BB	3	850,000	1.0	850,000	68,000
22	BBB	1	850,000	1.0	850,000	68,000
67	BBB	1	850,000	1.0	850,000	68,000
13	A	3	850,000	0.5	425,000	34,000
23	A	1	850,000	0.5	425,000	34,000
93	B	1	184,000	1.5	276,000	22,080
5	CCC/C	1	174,000	1.5	261,000	20,880
96	B	3	163,000	1.5	244,500	19,560

2.4 Aggregated Breakdown by Rating Bucket

```
by_rating <- port_sa[, .(
  N      = .N,
  EAD    = sum(Loan.Volume),
  RW     = unique(RW),
  RWA    = sum(RWA),
  Capital = sum(Capital)
), by = Rating]

# Order by the canonical rating order, not alphabetical.
by_rating[, Rating := factor(Rating, levels = ratings_order)]
setorder(by_rating, Rating)
```



```
# Append totals row.
totals <- data.table(
  Rating = "Total",
  N       = sum(by_rating$N),
  EAD     = sum(by_rating$EAD),
  RW      = sum(by_rating$RWA) / sum(by_rating$EAD),
  RWA     = sum(by_rating$RWA),
  Capital = sum(by_rating$Capital)
)
by_rating_out <- rbind(
  by_rating[, .(Rating = as.character(Rating), N, EAD, RW, RWA, Capital)],
  totals
)
```

Table 10: Standardized Approach breakdown by rating bucket (EUR, RW in decimal).

Rating	N	EAD	RW	RWA	Capital
AAA	3	279,000	0.20	55,800	4,464
AA	8	1,410,000	0.20	282,000	22,560
A	17	3,430,000	0.50	1,715,000	137,200
BBB	30	4,821,000	1.00	4,821,000	385,680
BB	27	4,428,000	1.00	4,428,000	354,240
B	9	887,000	1.50	1,330,500	106,440
CCC/C	6	1,321,000	1.50	1,981,500	158,520
Total	100	16,576,000	0.88	14,613,800	1,169,104

The portfolio is concentrated in BBB (30 obligors, \$ 29% of EAD) and BB (27 obligors, \$ 27% of EAD), both of which carry the capital-neutral 100% weight. The eight AA and three AAA names absorb 20% weights, the seventeen A-rated obligors 50%, while the fifteen sub-investment-grade names (B and CCC/C combined) are penalised at 150%. No obligors are in default in the current book, so the SA exposure-weighted risk weight falls between the 75% (everyone BBB) and 150% (everyone CCC/C) extremes that would bracket this rating distribution.

2.5 Headline Capital Numbers

```
total_EAD <- sum(port_sa$Loan.Volume)
total_RWA <- sum(port_sa$RWA)
capital_req_SA <- 0.08 * total_RWA
avg_rw <- total_RWA / total_EAD
eff_cap_ratio <- capital_req_SA / total_EAD
```

Total exposure at default is EUR 16,576,000, total RWA is EUR 14,613,800, and the minimum capital requirement (8% of RWA) is EUR 1,169,104. The exposure-weighted average risk weight is 88.16%, and the effective capital-to-EAD ratio is 7.05%.

2.6 Interpretation and Limitations

The Standardized Approach delivers a minimum capital figure of roughly EUR 1.17 m, or 7.05% of total exposure. That number is mechanically driven by the rating distribution and a coarse step-function of risk weights; it ignores three economically important features of the portfolio. *First*, it makes no use of the empirical PDs we just estimated in Task 1: a BBB and a BB obligor receive identical 100% weights despite their Lando–Skodeberg PDs differing by a factor of more than three. *Second*, it does not account for concentration risk; the largest single exposure attracts exactly the same per-euro weight as the smallest. *Third*, it implicitly assumes independence — there is no correlation parameter and therefore no notion of tail or unexpected loss. Tasks 3 (IRB), 4 (2-factor Merton with a common market factor) and 5 (2-factor Merton with industry factors) progressively relax these assumptions, and the resulting capital figures are compared in Section 6.

3 Task 3 — Basel IRB Approach

Author: Balazs Limpek

3.1 Case-Study Task

3.2 Implementation

3.3 Results

4 Task 4 — 2-Factor Merton with Common Market Factor

Author: Balazs Limpek

4.1 Case-Study Task

4.2 Implementation

4.3 Results

5 Task 5 — 2-Factor Merton with Industry Factors

Author: Caterina Piacentini

5.1 Case-Study Task

5.2 Implementation

5.3 Results

6 Comparison and Conclusions

Author: Caterina Piacentini

6.1 Capital Requirements Across Methods

6.2 Discussion

6.3 Limitations and Caveats